Unit 3 Pre-Test

Systems of Equations & Inequalities

Standards assessed: NC.M1.A-CED.3 · A-REI.5 · A-REI.6 · A-REI.10 · A-REI.11 · A-REI.12

Name: _____ **Date:** _____ **Period:** _____

SCORE 3.0 Basic Understanding

NC.M1.A-REI.6 · NC.M1.A-REI.10 · NC.M1.A-REI.12 Check solutions and solve simple systems.

• *I can tell if a point is a solution, read a solution from a graph, and solve simple systems.*

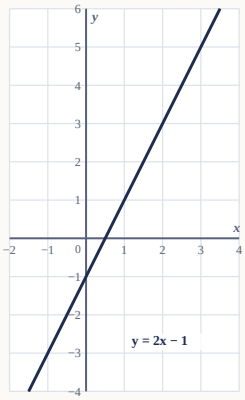
Answer each question. Use the picture to help you.

How to type your answers

point `(2, 5)` number `-3` yes / no `yes` inequality `y<=-x+4`

1. *Solution?*

Is the point (2, 3) a solution of y = 2x − 1? Write yes or no.

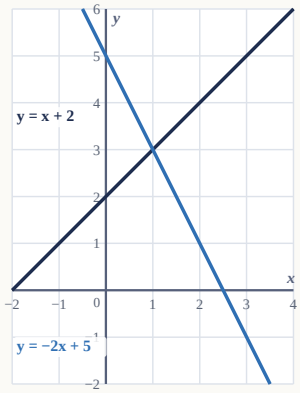


Answer: _____

`answer: yes`

2. *Solve by graphing*

Use the graph. What is the solution of the system?



Answer: _____

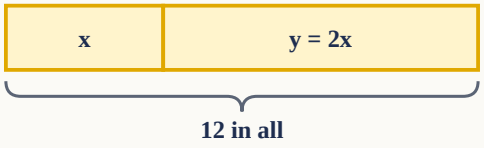
`answer: (1, 3)`

3. *Substitution*

Solve the system.

y = 2x

x + y = 12

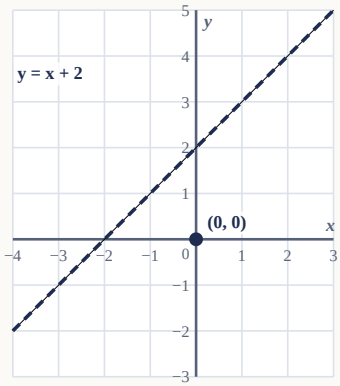


Answer:

answer: (4, 8)

4. *Inequality*

Is (0, 0) a solution of y > x + 2? Write yes or no.



Answer:

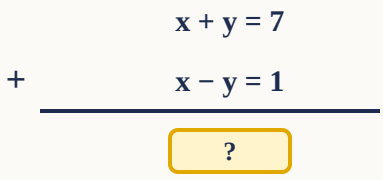
answer: no

5. *Elimination*

Solve the system.

x + y = 7

x − y = 1

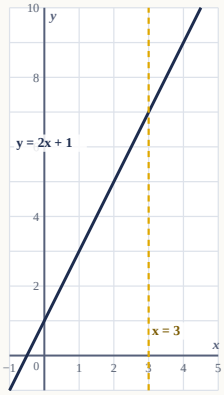


Answer:

answer: (4, 3)

6. *Point on a line*

The point (3, k) is on the line y = 2x + 1. Find k.



k =

answer: k = 7

SCORE **4.0** **Target Content**

NC.M1.A-REI.5 · NC.M1.A-REI.6 · NC.M1.A-REI.12 · NC.M1.A-CED.3 Solve systems by substitution and elimination and write systems from words.

• *I can solve systems with substitution or elimination, tell how many solutions there are, and write a system for a situation.*

Solve each problem. Show your work on paper.

7. *Elimination*

Solve the system.

$2x + 3y = 12$

$x - 3y = -3$

$$\begin{array}{rr} & 2x + 3y = 12 \\ + & x - 3y = -3 \\ \hline & \boxed{?} \end{array}$$

Answer:

answer: (3, 2)

8. *Substitution*

Solve the system.

$y = 3x - 5$

$2x + y = 10$

$$\boxed{2x} \quad \boxed{+} \quad \boxed{y} \quad \boxed{= 10}$$

$$y = 3x - 5$$

Answer:

answer: (3, 4)

9. *How many solutions?*

How many solutions does this system have?

$y = 2x + 3$

$y = 2x - 1$

line	slope	y-int
A	?	?
B	?	?

Answer:

answer: no solution (parallel lines)

10. *Write a system*

Adult tickets cost $8. Student tickets cost $5. 20 tickets sold for $130 in all. How many **adult** tickets were sold?

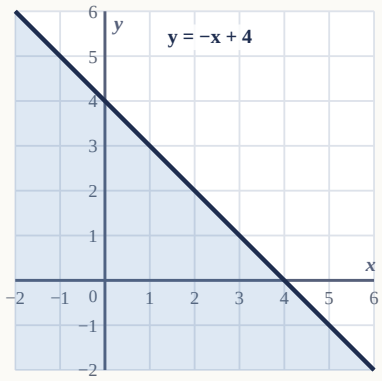
	adult	student	total
tickets	a	s	20
dollars	8a	5s	130

adult =

answer: 10 adult tickets

11. *Read the graph*

Write the inequality shown by the graph. (The line is solid. The shading is below.)



Answer:

answer: $y \le -x + 4$

12. *Elimination step*

$x + 2y = 7$

$3x - y = 7$

Multiply the second equation by 2, then add the equations. What new equation do you get?

$$\begin{array}{rl} & \mathbf{x + 2y = 7} \\ + & \mathbf{3x - y = 7} \quad \times 2 \\ \hline & \boxed{?} \end{array}$$

Answer:

answer: $7x = 21$

SCORE **5.0** **Advanced Content**

NC.M1.A-REI.6 · NC.M1.A-REI.11 · NC.M1.A-REI.12 · NC.M1.A-CED.3 Solve harder systems and use constraints.

• *I can solve any linear system, test systems of inequalities, and use constraints in real situations.*

Solve each problem. Some answers are words.

13. *How many solutions?*

How many solutions does this system have?

$2x - y = 3$

$4x - 2y = 6$

line	slope	y-int
A	?	?
B	?	?

Answer:

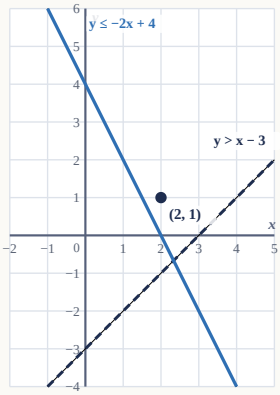
answer: infinitely many solutions (same line)

14. *System of inequalities*

Is (2, 1) a solution of the system?

$y > x - 3$

$y \le -2x + 4$



Answer:

answer: no

15. *Constraint*

Shirts cost $12. Hats cost $8. Ana can spend **at most** $96. She buys 5 shirts. What is the greatest number of hats she can buy?

5 shirts × $12	? hats × $8

at most $96

hats =

answer: 4 hats

16. *Elimination*

Solve the system.

$3x + 2y = 16$

$2x - 3y = -11$

$$\begin{array}{rl} & 3x + 2y = 16 \quad \times\ ? \\ + & 2x - 3y = -11 \quad \times\ ? \\ \hline & ? \end{array}$$

Answer:

answer: (2, 5)

17. *f(x) = g(x)*

f(x) = 2x + 1 and g(x) = −x + 7. For what value of x is f(x) = g(x)?

x	0	1	2	3
f(x)	1	3	?	?
g(x)	7	6	?	?

x = _____

answer: x = 2

18. *Write a system*

Two numbers add to 30. Their difference is 8. What is the **larger** number?

larger = small + 8



Answer: _____

answer: 19